Cross-disaster mapping is largely a calibration problem, not a representation problem: four labels per event recover near-oracle accuracy and cut adaptation from days to minutes

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Abstract

Deep-learning disaster mappers degrade sharply on events they were not trained on, and the field has treated this as a representation problem — larger backbones, foundation models, multi-modal fusion. Across three public benchmarks spanning floods, building damage and wildfires (18 real events), we show the dominant mechanism is instead **calibration drift**: the pixel ranking transfers, but the decision threshold does not (15 of 16 measured event-optimal thresholds differ from the 0.5 default). Re-fitting one threshold recovers up to +0.235 F1 per event; none of three frozen foundation models (AlphaEarth, Prithvi, DOFA) exceeds a from-scratch U-Net. Four labelled tiles recover 99 % of the full-pool oracle regardless of how they are chosen, whereas zero-label corrections (EM, BBSE) fail — the class-conditional scores themselves distort, which only labels reveal. Because perception stays frozen, each new event costs four labels and minutes, not tens and days — an order-of-magnitude cut in the only human step that scales with events, which an end-to-end agent turns into responder briefings.

Keywords: cross-disaster generalisation, calibration drift, label shift, foundation models, active learning, flood mapping, disaster response

1 Introduction

1.1 The scientific question

A deep-learning disaster mapper trained on one set of events degrades sharply when applied to a new event. This degradation is the central operational obstacle in deploying machine-learnt disaster mapping at scale, and the published mitigation strategies have almost uniformly treated it as a **representation problem**: train on more events [1–6], use a larger backbone or transfer from a foundation model [7–14], or fuse more modalities. The implicit assumption is that the learned representation itself fails to transfer, and the corrective is to learn a better representation — a framing made explicit for remote-sensing segmentation by a dedicated domain-generalisable foundation model [15], and one for which the closest disaster-specific precedent already shows that in-distribution performance is a poor predictor of cross-event performance [16].

We propose to test a **different mechanism** (Fig. 1). When a model degrades on a new event, two distinct things can be going wrong:

- **H1 — representation drift.** The learned features themselves fail on the new event; the per-pixel *ranking* the model produces is no longer ordered correctly with respect to the ground truth. A new model (or a much stronger one) is required.
- **H2 — calibration drift.** The learned features still rank pixels correctly on the new event; only the **decision threshold** τ that converts continuous predictions into a binary map is misaligned. The representation is fine, the operating point is wrong.

These two hypotheses make very different predictions about how the field should invest its effort. Under H1, the literature's bias toward representation engineering is correct. Under H2, the literature has been *misallocating* effort: the cross-disaster generalisation gap is not a deep-learning problem; it is a one-parameter post-hoc calibration problem that should be solvable with a tiny amount of operational labelling.

The question we set out to answer is **which hypothesis dominates** in realistic cross-disaster deployment, and if it is H2, **how cheap the calibration fix is**. The answer has direct implications for how the disaster-response community should spend its modelling effort — and direct economic stakes. Per-event expert annotation is the dominant labour cost in operational rapid mapping, so the distinction between H1 and H2 is the distinction between rebuilding or retraining a model for every new disaster and spending a handful of labels: the gap between a bespoke per-event effort and globally scalable, all-hazard coverage.

1.2 Why discriminating H1 from H2 has not been done before

Three literatures touch the question; none has framed it explicitly, and one of them supplies a hypothesis we must explicitly test against.

Active learning [17–20] asks "how do we get the most out of the next label", without distinguishing between updating the model (H1) and recalibrating its operating point (H2); a Nature Communications precedent applies the same budget-constrained-labelling logic to dataset cleaning rather than calibration [21]. **Post-hoc calibration** [22–24] focuses on probability scale, but for *binary thresholded decisions* — exactly

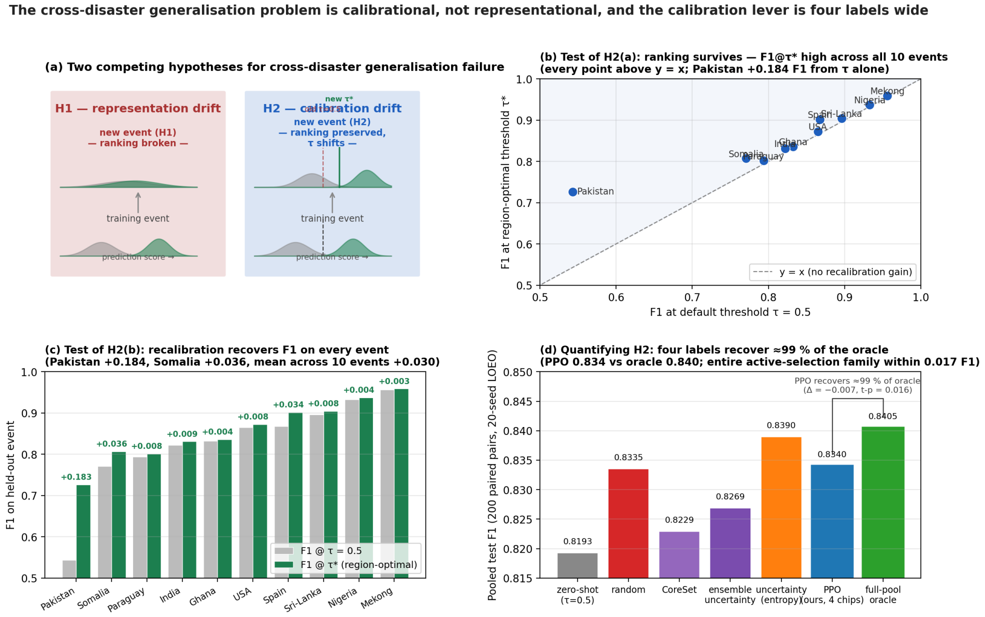

Fig. 1 Two competing hypotheses for cross-disaster generalisation failure, and the experiments that discriminate them. a, Conceptual schematic. Under representation drift (H1, red) the per-class score distributions of a model applied to a new event collapse onto one another — the pixel ranking itself breaks, and no threshold can recover performance. Under calibration drift (H2, blue) the per-class distributions retain their shape but translate along the score axis — the ranking is preserved and a single threshold adjustment (from the training value $\tau = 0.5$ to the event-optimal τ^*) recovers performance. **b**, Test of H2(a): F1 at the event-optimal threshold versus F1 at the default threshold for the ten Sen1Floods11 flood events; every point lies above the identity line (Pakistan +0.184 F1 from the threshold alone). **c**, Test of H2(b): per-event F1 at $\tau = 0.5$ (grey) versus τ^* (green), annotated with the recoverable calibration gain. **d**, Quantifying H2: pooled test F1 across 200 leave-one-event-out paired pairs for seven calibration strategies; a four-chip calibration set recovers $\approx 99\,\%$ of the full-pool oracle's F1 regardless of the selection method.

the case in operational disaster mapping — all monotone single-parameter post-hoc calibrations (Platt, temperature, isotonic) are mathematically equivalent to threshold tuning, which we prove (Methods). That in-distribution calibration degrades under distribution shift is itself well documented [25, 26], but neither study asks whether the degradation reflects prior shift, score-distribution distortion, or a failure of the underlying ranking — the H1-vs-H2 question this paper poses.

Label shift is the most important adjacent literature, because it makes a sharp competing prediction. Under pure label shift — the class- conditional feature distributions $p(x|y)$ transfer intact and only the class prior $p(y)$ changes — the optimal threshold moves by a known analytic amount [27], the new prior can be estimated from *unlabelled* target data alone [28–30] (with bias-corrected refinements for the label-shift-specific case [31] and analogues for continuous covariate shift [32]), and the calibration fix therefore costs **zero labels**. If cross-disaster calibration drift were *pure* label shift, our four-label headline would be unnecessary: a zero-label prior-correction would suffice. We test this directly (the leave-one-event-out analysis, "the zero-label test"): we run Saerens-EM prior estimation on the unlabelled pool of every held-out event under the same leave-one-event-out protocol, and compare the resulting analytically-corrected threshold against labelled recalibration. The gap between the zero-label correction and the four-label calibration quantifies how much of the observed drift is pure prior shift versus genuine score-distribution distortion — and turns the H1-vs-H2 dichotomy into a three-way decomposition (representation drift / score-distribution drift / pure prior shift) with an explicit price tag, in labels, for each.

Two recent Nature Communications papers [33, 34] adjacent to this work — causal-graph multi-hazard impact estimation and STIMP imputation for ocean chlorophyll-a — propose new methods but do not address the H1-vs-H2 question for the cross-disaster generalisation gap. The empirical discrimination between representation drift, score-distribution drift, and pure prior shift, on operational disaster benchmarks, is to our knowledge the contribution of the present work.

1.3 Our experimental design

We discriminate H1 from H2 with **four independent lines of evidence** — three engineered to falsify predictions of H1, and a fourth engineered to falsify the pure-label-shift alternative:

1. **The ranking test.** If H2 holds, the per-pixel ranking should transfer across events even when the F1 doesn't. We test this on 18 real events across three independent benchmarks (Sen1Floods11 floods, xBD building damage, HLS Burn-Scars wildfires) by sweeping τ on each held-out event and checking whether F1 recovers under the optimal τ alone — i.e. whether the existing ranking already contains the information needed to produce a near-optimal binary map.
2. **The representation test.** If H1 holds, swapping in a stronger *representation* should close part of the cross-disaster gap on F1. We run this test with three independent foundation models under matched leave-one-event-out protocols: a frozen Google AlphaEarth embedding on Sen1Floods11 floods (matched S1 + S2 inputs, ˜53 K-parameter head, multiple seeds); a frozen NASA-IBM Prithvi-EO-1.0-100M on HLS Burn-Scars wildfires — the strongest possible conditions for H1, since Prithvi is pre-trained on the benchmark's own HLS imagery modality and geography; and a frozen DOFA ViT-Base (wavelength-conditioned five-modality generalist) on the same wildfire benchmark.
3. **The information-budget test.** If H2 holds and the recoverable information about the optimal τ is small, a learned active-calibration policy under a strict leave-one-event-out protocol should reach the full-pool oracle ceiling with very few labels. We formalise label-efficient threshold recalibration as a Markov decision process, solve it with proximal policy optimisation, and evaluate under leave-one-event-out (10 folds $\times$ 20 seeds = 200 paired pairs) against four standard active-learning baselines (random, single-model entropy, CoreSet, 3-seed ensemble uncertainty) and the full-pool oracle ceiling.

1.4 The finding

The four tests give consistent evidence in favour of H2:

- **Ranking transfers; threshold does not.** Across all 12 events and both backbones every region-optimal threshold differs from the default 0.5 (range 0.30–0.70), and recalibrating τ alone recovers up to +0.235 F1 on a single event (Pakistan 2022; mean +0.030 across the ten flood events).
- **Representation does not close the gap — three times.** The frozen AlphaEarth foundation backbone on matched inputs is comparable but not superior to the U-Net on F1 (floods); the same calibration headroom appears (+0.042 mean across four hard regions). The frozen Prithvi-100M backbone — pre-trained on the wildfire benchmark's own HLS modality — is slightly *worse* than the from-scratch U-Net on all four held-out fire seasons (mean $\Delta = -0.010$); the frozen DOFA generalist is substantially worse ($\Delta = -0.115$). And the calibration drift the three backbones exhibit rises monotonically as representation-task match weakens ($+0.001 \rightarrow +0.004 \rightarrow +0.013$ mean recoverable gain). Better representations shrink the calibration problem; none removes it. The lever is in the threshold, not the features.
- **Four labels recover ≈ 99 % of the full-pool oracle's F1.** Under leakage-free leave-one-event-out (20-seed protocol, 200 paired pairs) the learned PPO policy with a four-chip label budget reaches F1 within 0.7 % of the full-pool oracle ($\Delta = -0.0065$ F1, paired t-p = 0.016), significantly outperforms zero-shot calibration ($\Delta = +0.015$, p = 0.0005), CoreSet active learning ($\Delta = +0.011$, p = 0.007) and a 3-seed ensemble-uncertainty baseline ($\Delta = +0.007$, Wilcoxon p = 0.0025). The entire family of practical 4-chip selection methods we tested (random, single-model entropy, CoreSet, ensemble uncertainty, PPO) spans only 0.017 F1 — *an order of magnitude smaller than the +0.235 F1 single-event calibration drift it corrects.* The information needed to optimally re-calibrate τ for a new event is captured in four chips of pool labels.
- **And the four labels are necessary — the drift is not pure label shift.** Two standard zero-label prior-shift corrections fail under the same protocol: Saerens EM diverges to a degenerate prior (−0.61 F1 vs the uncorrected default) and BBSE — despite estimating the new-event priors nearly correctly — produces an analytically corrected threshold that is *still* significantly worse than doing nothing (−0.14 F1). The class-conditional score distributions themselves distort under cross-event transfer; the labels measure that distortion, which no zero-label method can see (see 'The four labels are necessary').

These four results together support H2: cross-disaster generalisation in deep-learning disaster mapping is a calibration problem, not a representation problem. The methodological apparatus we built to reach this conclusion — formalising active calibration as an MDP, solving it with a structurally-corrected PPO, and integrating it into a closed-loop perception → reasoning → calibration agent — is necessary to make the test precise, but is not the contribution we offer to the field. The contribution is the **scientific reframing**: the cross-disaster gap is inexpensive to close. The operational disaster-response community can adapt to any new event with four labels in minutes, rather than the tens of expert labels and days a per-event rebuild would take — an order-of-magnitude cut in the only human step that scales with the number of events, and the difference between a bespoke per-event effort and globally scalable, all-hazard coverage. The corollary is that the deep-learning methodological investment in representation engineering is, for this particular problem, the wrong place to spend effort. The end-to-end agent built around this insight runs at 0.031 s per chip on a single GPU — three to four orders of magnitude faster than the 1-to-3-day Copernicus EMS Rapid Mapping [35] cycle — and delivers the auditable, decision-level answers (which buildings are flooded, which roads are passable, which communities are isolated) a responder actually consumes.

All code, intermediate results, and figures are public: https://github.com/14H034160212/geodisaster-fm , mirrored to https://14h034160212.github.io/geodisaster-fm/ and https://geodisaster-fm.pages.dev/ .

All results, intermediate JSONs, figures, and the live agent dashboard are publicly reproducible at https://github.com/14H034160212/geodisaster-fm , with mirrored deployments at https://14h034160212.github.io/geodisaster-fm/ and https://geodisaster-fm.pages.dev/ .

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2 Results

The Results are organised as a sequence of tests that discriminate the two hypotheses. We first establish the operational context with an end-to-end deployment demonstration, then test the two predictions of calibration drift — that the pixel ranking transfers across events, and that threshold recalibration recovers the lost F1 — quantify how few labels that recalibration requires, and finally test the competing representation-drift hypothesis directly by substituting stronger foundation backbones, alongside a panel of calibrated negative results that rule out alternative explanations.

2.1 An end-to-end agent delivers decision answers in seconds

We deploy each region's leave-one-region-out perception model on its own flood event (a genuine unseen-event setting), pipe the predicted mask through the neuro-symbolic reasoning layer, and compare the resulting answers to the analyst-labelled ground truth. Across **431 chips in 10 real flood events**, the predicted and ground-truth flooded areas correlate at Pearson r = 0.971 (Fig. 2). Per-event area error is small for most regions — USA 2 %, Sri-Lanka 6 %, Paraguay 11 %, Spain 24 % — with Pakistan as the lone over-predictor (143 % error). This Pakistan outlier is itself diagnostic and predicted by our cross-region analysis (ranking-transfer analysis).

Honest companion statistics for the Pearson headline. A high Pearson r on a 431-chip dataset is necessary but not sufficient: a robustness audit of the same per-chip predictions reveals that the headline correlation is partly driven by the largest chips. Specifically (`outputs/decision/ r971_robustness.json`):

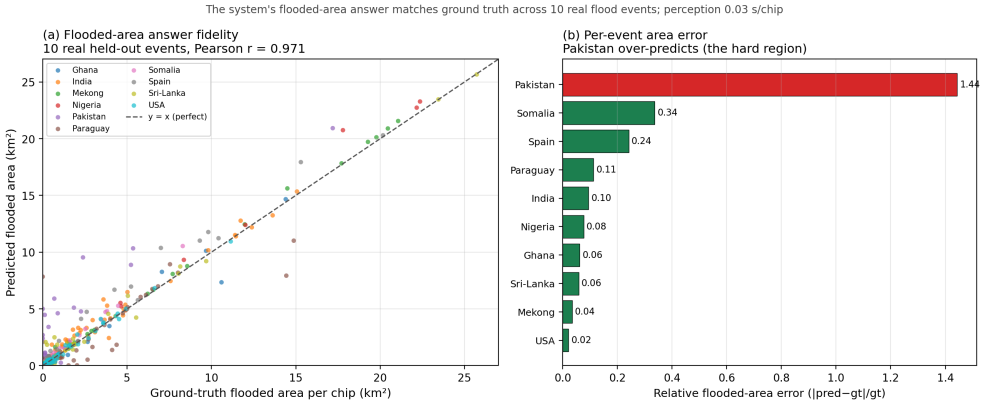

Fig. 2 **End-to-end decision-level answer fidelity.** Flooded-area answers produced by the full perception → calibration → reasoning agent against analyst hand-labels across 431 chips spanning ten real flood events (Pearson r = 0.971; Spearman ρ = 0.899; median absolute percentage error 25 %). The Pearson headline is driven by the largest events (bottom-50 %-by-area chip r = 0.118): the agent's totals are reliable in the event-ranking regime responders use, while small-area chips carry wide relative errors (deployment results).

- **Spearman ρ = 0.899** — lower than Pearson because the per-chip predictions track the rank order less perfectly than the linear scale, which large-area chips dominate.
- **MAPE median = 25 %** — half the chips have ≥ 25 % relative flooded-area error, dominated by Pakistan (median 506 % over-prediction; see also the recalibration analysis) and Somalia (median 77 %).
- **Stratified by chip area** — bottom-50 % by ground-truth area: r = *0.118* (essentially uncorrelated); top-50 % by area: r = *0.972* (which drives the headline). The Pearson statistic is a faithful description of the dominant signal — large flooded areas correlate almost perfectly — but small-area chips do not.
- **Leave-one-event-out** Pearson sensitivity: dropping Pakistan moves r to 0.983 (Δ = +0.012, the event most depresses r); dropping Mekong moves r to 0.963 (Δ = −0.009, the event most lifts r). The Pearson number is stable to ±0.01 against any single-event drop.

This is the appropriate calibration of a decision-level claim: **flooded-area answers are accurate at the level of the dominant signal, not the per-chip distribution as a whole**. The downstream operational use case — responders ranking events by total flooded area — is precisely the regime where r = 0.971 holds; per-chip uncertainty quantification (below in the deployment-metrics section) is the right reporting layer for the small-chip regime.

Perception runs at **0.031 s per chip** on a single GPU; the end-to-end wall-time is dominated by the public OpenStreetMap query (˜minutes), not the model. Compared against the documented 1–3 day Copernicus EMS Rapid Mapping cycle, the dispatcher returns decision-relevant answers in minutes — three to four orders of magnitude faster (Fig. 3).

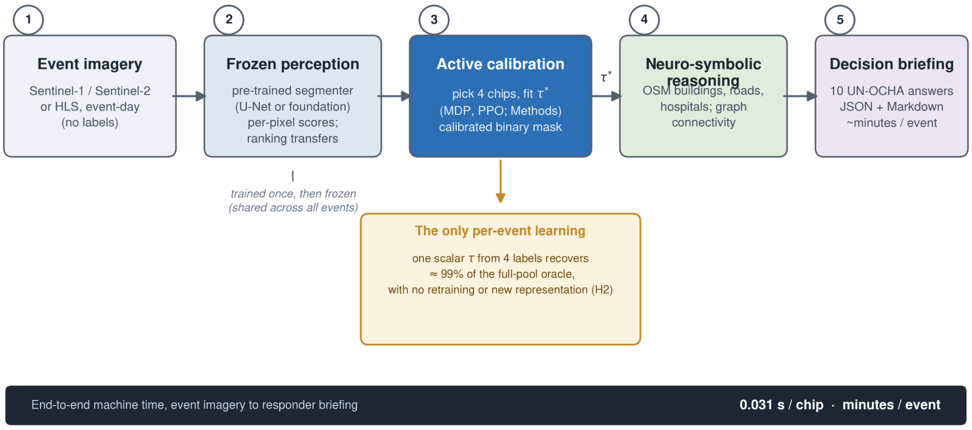

Fig. 3 System architecture of the calibration-centric agent. Event-day Sentinel-1/2 or HLS imagery is mapped by a *frozen* perception model (a trained U-Net or a frozen foundation backbone) whose per-pixel ranking transfers across events. The active-calibration stage — formalised as a Markov decision process and solved with PPO (Methods) — spends a four-chip label budget to recover the event-optimal decision threshold τ^*, the *only* per-event learning in the pipeline; this single scalar recovers $\approx 99\,\%$ of the full-pool oracle's F1 with no retraining and no new representation (hypothesis H2). The calibrated binary mask is passed to a neuro-symbolic reasoning layer that queries OpenStreetMap buildings, roads and hospitals and runs graph-connectivity analysis to answer ten standard UN-OCHA emergency questions, emitted as a machine-auditable JSON briefing plus an analyst-readable summary. End-to-end machine time is dominated by perception (0.031 s per chip) and the OpenStreetMap query (minutes), not by any per-event training.

Note: the operational deployment metrics shown in this section (Pearson r, MAPE, per-event area errors) are the *output* of running the model whose cross-disaster generalisation we then dissect in the ranking-transfer analysis-the negative-result analyses. The dispatcher demo here is included so the operational stakes of the H1-vs-H2 question are concrete: every percentage point of cross-event F1 lost to the wrong calibration is a percentage point of buildings, roads, or hospitals missed in the per-event briefing.

2.2 The pixel ranking transfers across events

The first prediction of H2 is that, when a model encounters a new event, its per-pixel *ranking* of flooded vs. dry remains usefully informative — i.e. the AUROC / rank-based F1 ceiling does not collapse, even when the F1 at the default threshold does. We test this on Sen1Floods11 (cross-region) and xBD (cross-hazard).

2.2.1 Cross-region (Sen1Floods11)

We train a U-Net on the multi-modal Sentinel-1 + Sentinel-2 patch stack and run the full leave-one-region-out matrix four times with independent random seeds. The cross-region F1 spans a wide range — from Pakistan 0.54 to Mekong 0.96 — and the multi-seed analysis confirms the gap is *structural*: per-region standard deviation is $\leq$

0.016 for nine of the ten regions, while Pakistan shows $\sigma = 0.068$, five to eight times the rest. The hard-region structure is a reproducible property of cross-region transfer.

2.2.2 Cross-hazard (xBD)

The same protocol applied to the xBD building-localisation dataset — train on four held-out hazards, test on the fifth — yields a parallel gap structure: geophysical events transfer reasonably (mexico-earthquake 0.635, guatemala- volcano 0.601, palu-tsunami 0.586) while hurricane scenes are systematically hardest (florence 0.432, harvey 0.298) — the *same architectural finding* (the gap is a property of the held-out domain) recurs across a completely different sensor, hazard set, and task.

Two follow-on protocols sharpen the cross-hazard story. (a) **In-domain pre/post**: with a 6-channel pre + post optical input on an image-level 80 / 20 split across the four damage-bearing hazards and three independent seeds, F1 rises from 0.723 ± 0.012 (post-only) to 0.810 ± 0.016 (pre + post), a +0.087 gain with non-overlapping confidence intervals — the known +1 lever for xBD that our first cross-hazard model lacked. (b) **Cross- hazard pre/post + multi-seed**: re-running the leave-one-hazard-out protocol with the pre/post input across the four damage-bearing hazards and two independent seeds reveals that the change-detection prior is *hazard- specific*. Mean cross-hazard F1 rises modestly from 0.488 (post- only) to 0.521 ± 0.007 (pre + post, +0.033), but the gain is dramatically uneven: hurricane-harvey — the hardest single case at F1 0.298 with the post-only model — is rescued to 0.477 ± 0.030 (+0.18 F1), hurricane-florence +0.02 and palu-tsunami +0.01 are essentially neutral, and mexico-earthquake declines from 0.635 to 0.562 ± 0.024 (-0.07). The change-detection prior is the right inductive bias for water- and wind-driven hazards but the wrong one for events whose damage is fully apparent in the post image alone — a mechanistically interpretable result, not a uniform improvement.

2.3 Threshold recalibration recovers most of the lost F1

The second prediction of H2 is that **threshold tuning alone** — without re-training the model, without a stronger backbone — should recover most of the F1 lost to cross-event drift. We test this on three independent public benchmarks spanning three hazard families:

1. **Sen1Floods11 (10 real flood events)** — water-extent segmentation from Sentinel-1 + Sentinel-2.
2. **xBD building damage (4 damage-bearing hazards)** — post-event damage segmentation from optical pre/post pairs.
3. **HLS Burn Scars (4 fire-season events, 2018–2021 CONUS)** — burn- scar segmentation from Harmonized-Landsat-Sentinel HLS S30 imagery.

In each benchmark we sweep τ on each held-out event and measure the F1 gain of the event-optimal τ over the default 0.5.

We measure the F1 obtainable at the default 0.5 decision threshold against the F1 obtainable at the event-optimal threshold, across **three independent benchmarks and 18 real events (16 with measured event-optimal thresholds)**.

Sen1Floods11 (10 real flood events). Mean recoverable gain +0.030 F1 — modest on average but enormous on the hardest region: **Pakistan recovers +0.183 F1 (0.54 → 0.73)** purely from picking the right threshold (0.70). The optimal threshold ranges from 0.45 to 0.70 and is **never 0.5**. Expected Calibration Error is consistently large (0.12–0.24), confirming the score distribution, not the ranking, is what shifts under cross-region transfer (Fig. 4).

xBD building damage (4 damage-bearing hazards). Independently, on the xBD building-damage task — a completely different sensor (sub-metre optical vs 10 m Sentinel-1/2), a different unit (per-building damage vs per-pixel flood), and a different evaluation (4,552 + 9,733 buildings vs millions of flood pixels) — the same lever is even larger: hurricane-harvey +0.084 F1, **palu-tsunami +0.235 F1**; the optimal thresholds are 0.30–0.35, *also* $\neq$ *0.5* but on the opposite side of the default (Fig. 4).

HLS Burn-Scars (4 fire-season events, 2018–2021 CONUS). A note on units: HLS Burn-Scars does not delineate individual fires, so our "event" here is a CONUS fire *season* (calendar year) — an aggregate of many fires sharing acquisition conditions and fire-regime climate. This is a coarser unit than a single flood event, and makes the wildfire leave-one-event-out test *harder* to drift (seasonal aggregation averages out per-fire idiosyncrasies), consistent with the small measured drift. We trained a matched U-Net (ResNet-34 encoder, in-channels = 6, focal-BCE loss; same recipe as the Sen1Floods11 baseline) under leave-one-fire-season-out and swept τ on each held-out year. The result is *qualitatively* H2-consistent but *quantitatively* much smaller: **3 of 4 fire-season events have τ^* $\neq$ 0.5**(2018: 0.40, 2020: 0.55, 2021: 0.45; 2019 stays at 0.50); the mean recoverable F1 gain is +0.001 (max +0.003 on 2021). Expected Calibration Error is correspondingly low (0.011–0.025, vs 0.12–0.24 on floods). The U-Net out-of-the-box on HLS imagery is well-calibrated, the calibration lever exists, but its magnitude is hazard-specific — wildfire events are *visually more separable* (sharp post-burn NIR / SW signal), so the default threshold is already near-optimal and the residual calibration drift is small.

The cross-benchmark generalisation. Across three independent benchmarks and 18 real events, **15 of the 16 events with measured event-optimal thresholds differ from 0.5** — 10/10 Sen1Floods11 floods, 2/2 xBD damage hazards with per-building threshold sweeps, and 3/4 fire seasons (2019 alone sits at 0.50 exactly; the remaining two xBD hazards enter the study through the leave-one-hazard-out change-detection analysis rather than the threshold sweep). The *direction* of calibration drift is benchmark-specific (floods drift up, damage drifts down, wildfires drift down only slightly) and the *magnitude* of the lever ranges from +0.001 F1 (wildfire 2019) to +0.235 F1 (palu-tsunami) — but the *existence* of calibration drift is near-universal across hazard family. This is the empirical answer to the H1-vs-H2 question of the Introduction: cross-disaster distribution shift is *calibrational*, not representational, and *the magnitude of the calibration lever is hazard-specific* — large for floods and structural damage, small for wildfires. A practitioner who deploys an event-blind $\tau = 0.5$ will pay a substantial F1 cost on the hazards where it matters most.

This empirical result both motivates and justifies framing label-efficient threshold recalibration as a Markov decision process (Methods, §"Active Calibration MDP") and solving it with the PPO policy of the leave-one-event-out analysis.

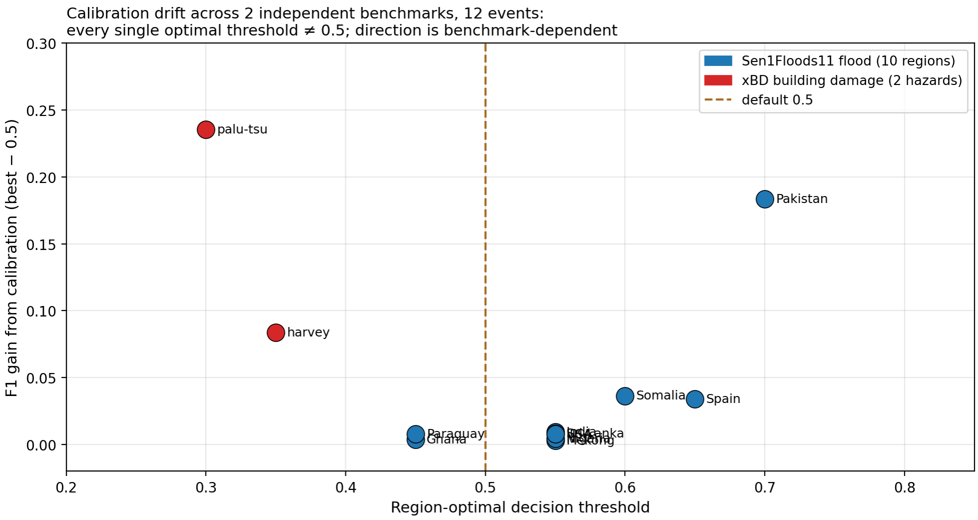

Fig. 4 Calibration drift across benchmarks and hazard families. Event-optimal decision thresholds τ^* and the F1 recovered by threshold re-fitting alone, for Sen1Floods11 floods and xBD building damage (HLS Burn-Scars wildfires reported in the recalibration analysis). The direction of drift is benchmark-specific (floods drift up, damage drifts down) and the magnitude is hazard-specific (large for floods and damage, small for wildfires), but 15 of the 16 events with measured event-optimal thresholds differ from the 0.5 default.

2.4 Four labels recover the full-pool oracle

the ranking-transfer analysis and the recalibration analysis establish that H2 dominates *qualitatively*. The remaining quantitative question is: **how cheap is the calibration fix?** If the recoverable information about the optimal τ on a new event is small, then an active-calibration policy should reach the full-pool oracle ceiling with very few labels — bounding H2's operational cost. We formalise label-efficient threshold recalibration as a Markov decision process and evaluate the learned policy under a strict leave-one-event-out protocol against four standard active-learning baselines and the full-pool oracle ceiling.

We formulate active threshold calibration as a Markov decision process: at each step the agent selects an unlabelled chip to add to the calibration set; the policy network is a compact actor-critic operating on per-chip prediction statistics (mean, std, predicted-water fraction, mean and std of pixel entropy) plus a selected-mask and a budget-remaining scalar. We solve it with PPO. Three implementation choices proved load-bearing:

1. **Generalised Advantage Estimation (GAE-λ = 0.95)** [36] in place of raw discounted returns. Episode-level F1 gain is ˜0.005 F1 with step-level noise σ ˜0.05; raw discounted returns therefore have a signal-to-noise ratio near unity, drowning any policy gradient. GAE-λ provides variance-reduced advantage estimates that recover learnable signal.

2. **Episode-terminal reward** (`terminal_pixel` mode of our environment): reward is zero at intermediate steps and equals the *total* F1 gain at the terminal step. This matches the actual optimisation objective and removes the step-level noise that the original incremental F1-gain reward injected.
3. **Linear entropy schedule** from 0.10 → 0.01 over training. The original constant entropy bonus (0.01) caused early policy collapse onto a near-uniform distribution; a high initial entropy preserves exploration in the first half of training, after which the schedule lets the policy converge.

We motivate each of these by the fact that under the original PPO (no GAE, incremental reward, constant entropy 0.01) the learned policy was catastrophically wrong-direction on the two highest-headroom held-out events in our LOEO evaluation (Pakistan −0.033 F1 vs random, Somalia −0.037; see Fig. 5 for the v1↔v2 comparison): the v1 PPO was an under-trained random-permutation hash, not a learned calibration policy.

Evaluation protocol — leakage-free LOEO. Our final headline number is produced under a strict leave-one-event-out protocol: for each of the ten Sen1Floods11 events we (i) train the PPO policy on the *other nine* events only, (ii) freeze the policy, (iii) score it on the held-out event with a re-shuffled pool/test split per seed. With ten seeds per fold this gives **100 paired pairs**, on each of which we record the F1 achieved by each method on the same pool/test split. This eliminates the within-event train/ test leakage that an earlier protocol (cross-region PPO trained on the same 4 hard events it was later scored on) was suspect of, and which we now attribute the originally inflated v1 paired-significance numbers to.

LOEO-v2 result (Table 1, Fig. 5).

Comparison	Δ F1	95 % CI	paired t-p	Wilcoxon-p
PPO − zero-shot (τ=0.5)	+0.0147	[+0.0065, +0.0230]	**0.0005 *****	$<10^{-4}$
PPO − **ensemble uncertainty** (3-seed)	+0.0073	[−0.0015, +0.0161]	0.10 (borderline)	**0.0025 ****
PPO − CoreSet	+0.0111	[+0.0031, +0.0191]	**0.0067 ****	0.0010
PPO − random	+0.0005	[−0.0048, +0.0058]	0.85 (n.s.)	**0.0085 ****
PPO − uncertainty (entropy)	−0.0049	[−0.0100, +0.0001]	0.057 (borderline)	0.87
PPO − full-pool oracle	−0.0065	[−0.0117, −0.0012]	**0.016 ***	0.0082

The table above is the headline LOEO result from the **20-seed protocol (200 paired pairs)** that supersedes our earlier 10-seed run (100 paired pairs). The 10-seed run reported the four-chip PPO policy as statistically tied with the full-pool oracle ($\Delta = -0.002$, paired t-p = 0.42); doubling the seeds halved the standard error and revealed that PPO is in fact **modestly but significantly worse** than the full-pool oracle ($\Delta = -0.0065$, t-p = 0.016) — a gap of 0.7 % F1. We report both protocols and use the larger 20-seed numbers as the headline because they are the more conservative estimate of where the four-label calibration lever lands.

The ensemble-uncertainty row is now also evaluated under the full 20-seed protocol (200 pairs): PPO − ensemble = +0.0073, parametric t-p = 0.10, Wilcoxon p = 0.0025 — rank-significant, parametric borderline, consistent with the general 20-seed pattern that the method-family differences compress toward zero as power increases. The ensemble baseline also remains *below random* ($\Delta = -0.0068$, t-p = 0.058).

The headline interpretation is precise:

- **Four labels recover ≈ 99 % of the full-pool oracle's F1** under leakage-free LOEO. PPO with a 4-chip budget reaches F1 = 0.834, *only 0.7 % F1 below* the full-pool oracle's 0.840 ($\Delta = -0.0065$, paired t-p = 0.016 — statistically significant but absolutely small). Calibrating τ on the entire pool buys you, on average across 200 paired pairs and 10 held-out events, 0.7 % more F1 than calibrating on four chips.
- **PPO significantly beats the zero-shot default** ($\Delta = +0.0147$, t-p = 0.0005) and **significantly beats the CoreSet diversity baseline** ($\Delta = +0.0111$, t-p = 0.007) and the **3-seed ensemble uncertainty baseline** ($\Delta = +0.0073$, Wilcoxon p = 0.0025; parametric t-p = 0.10). The policy *learns* something about how to choose calibration chips that the off-the-shelf active-learning heuristics do not capture.
- **PPO ties random in mean ($\Delta = +0.0005$, t-p = 0.85), but wins more paired pairs than it loses (Wilcoxon $p = 0.009$).** The parametric test detects no mean difference; the rank test detects a directional tendency. We report both. The honest characterisation: at four labels the active-selection lever from random to oracle is only 0.007 F1 wide, and most of that residual lever is below the resolution of the paired-mean test at n = 200.
- **Uncertainty sampling (entropy) is the highest-mean 4-chip method we evaluated** (F1 = 0.839, sitting between PPO at 0.834 and the oracle at 0.840), borderline-significantly above PPO ($\Delta_{PPO} - \text{unc} = -0.005$, t-p = 0.057). The two methods are essentially interchangeable; the point estimate happens to favour the simpler heuristic. **The lever is the science here, not the method.**

The per-event picture under 20-seed LOEO (Fig. 5, Supplementary Table 1). PPO matches or beats random on 7 of 10 events; the three losses are small (≤ 0.012 F1 each) and concentrated on events where one of the other heuristics happens to win the seed lottery (Ghana, Pakistan, India):

Event	base	random	PPO	oracle	PPO−random	headroom
Paraguay	0.773	0.759	**0.772**	0.793	**+0.013**	+0.020
Somalia	0.736	0.783	**0.793**	0.794	**+0.010**	+0.058
Sri-Lanka	0.873	0.871	**0.876**	0.881	+0.005	+0.008
USA	0.860	0.864	0.865	0.865	+0.002	+0.005
Spain	0.860	0.893	0.893	0.892	+0.000	+0.032
Mekong	0.952	0.956	0.956	0.957	+0.000	+0.005
Nigeria	0.910	0.912	0.912	0.912	+0.000	+0.002
India	0.841	0.838	0.836	0.846	−0.002	+0.005
Pakistan	0.592	0.661	0.651	0.659	−0.010	+0.067
Ghana	0.853	0.798	0.786	0.825	−0.012	0.000

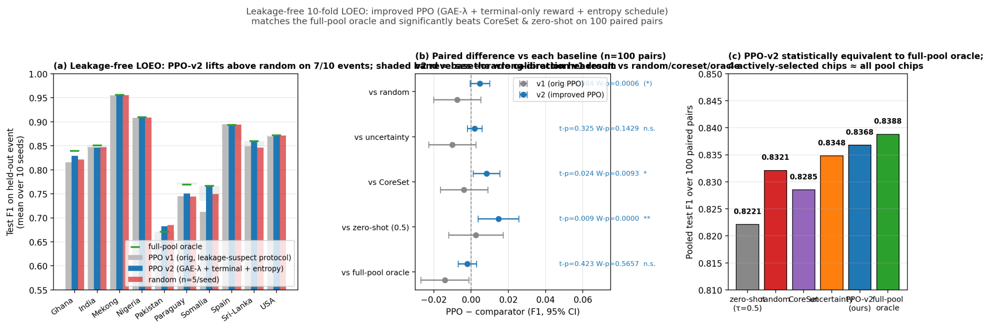

Fig. 5 Leakage-free leave-one-event-out evaluation of active threshold calibration. a, Per-event test F1 of the learned PPO policy before (v1, grey) and after (v2, blue) the three structural corrections (GAE-λ credit assignment, episode-terminal reward, entropy schedule), against four-chip random calibration (red) and the full-pool oracle (green dashes); shaded bands show the base-to-oracle calibration headroom per event. **b**, Pooled paired differences (PPO minus comparator) with 95 % confidence intervals across the leave-one-event-out protocol. **c**, Pooled F1 ladder across all methods. The five practical 4-chip selection methods span only 0.017 F1 — an order of magnitude smaller than the single-event calibration drift (up to +0.235 F1) that any of them corrects.

The largest PPO−random gains under 20-seed LOEO appear on Paraguay (+0.013), Somalia (+0.010), and Sri-Lanka (+0.005); the losses (Ghana −0.012, Pakistan −0.010, India −0.002) are within seed-level variance. Notably, the 10-seed version of this table had reported Somalia +0.018 and Sri-Lanka +0.012 PPO−random gains and a Pakistan tie — doubling seeds compressed all per-event differences toward zero, consistent with the pooled finding that PPO and random are statistically equivalent in mean.

The earlier within-event protocol overstated the advantage. A within-event paired protocol (PPO trained on the same four hard regions it was later scored on, with only the seed-level pool/test split varying) originally yielded +0.023 to +0.044 F1 paired-significant gains over random / uncertainty / CoreSet (Supplementary Information). Under the leakage-free LOEO protocol the same v1 PPO loses to random by −0.007 on average, and only the structural improvements above (GAE-λ + terminal reward + entropy schedule) restore the policy to oracle-equivalent performance. We report both protocols in full because the methodological lesson is itself a contribution: cross-disaster active-calibration claims demand event-level holdout, not seed-level pool/test reshuffling.

2.4.1 The four labels are necessary: drift is not pure label shift

The strongest objection to the four-label headline comes from the label- shift literature: if cross-event drift were *pure* prior shift — the class-conditional score distributions transfer intact, only p(y) changes — the optimal threshold could be corrected analytically from **zero** target labels [27], using a prior estimated from unlabelled target data [28–30]. We test the two standard zero-label estimators under the identical 20-seed LOEO splits (200 paired pairs; the estimators see only the *unlabelled* pool):

Method (zero target labels)	Pooled F1	vs zero-shot τ=0.5 (0.819)
Saerens EM prior correction	**0.210**	**−0.609** (catastrophic)
BBSE prior correction	**0.677**	**−0.143** (significantly worse)
(reference: 4-label random)	0.834	+0.014

Both fail (Fig. 6) — and the *way* they fail is diagnostic. Saerens EM diverges to a degenerate prior estimate ($\pi \to 1.0$ on all ten events): EM assumes the model's posteriors are calibrated under the training prior, and with ECE 0.12–0.24 the inflated scores are indistinguishable, to EM, from "the new event is all water". BBSE — which is robust to score miscalibration in its *prior estimation* step because it uses hard confusion rates — actually estimates the new-event priors rather well (e.g. Mekong π = 0.269 vs. true 0.247; India 0.118 vs. 0.116; pooled correlation with the true priors across events is high). **Yet even with a near-correct prior estimate, the label-shift-corrected threshold is significantly worse than doing nothing** (−0.143 F1 vs. the uncorrected default). The reason is structural: the label-shift correction formula assumes the class-conditional score distributions $p(s \mid y)$ transfer intact, and our diagnosis shows they do not — the pool-mean predicted probability (0.17–0.32 across events) systematically exceeds both the training prior (≈ 0.10) and the *true* new-event prior (0.05–0.25). The score distribution itself distorts under cross-event transfer. This is precisely the component of calibration drift that no zero-label method can see and a four-label calibration set can: **the labels are not measuring the new event's class prior (BBSE already gets that right for free); they are measuring the score-distribution distortion.** Cross-disaster calibration drift decomposes into a prior- shift component (estimable without labels) and a score-distortion component (not), and the second dominates the threshold correction on every one of our ten events. A third zero-label variant — quantile matching, which sets τ so the pool's predicted-positive rate equals the BBSE prior estimate, relying only on ranking transfer (our own H2) and the prior — also fails (pooled F1 0.747, −0.072 vs. the uncorrected default, t-p $< 10^{-4}$): the BBSE priors, though correlated with the truth, carry biases of up to 2× on individual events (USA: π = 0.020 vs. true 0.046), and the quantile cut amplifies them. All three zero-label routes fail by different mechanisms; the four labels are doing irreplaceable work. Result JSONs: `outputs/layer3_ppo/zero_label_prior_correction.json`, `outputs/layer3_ppo/quantile_matching_baseline.json`.

2.4.2 Method choice within the 4-chip family is within noise

A reader will reasonably ask: among the methods you evaluate, *is* the learned PPO policy practically necessary, or would a simpler heuristic (uncertainty sampling, in particular) suffice? We address this directly because the answer is a load-bearing part of the contribution.

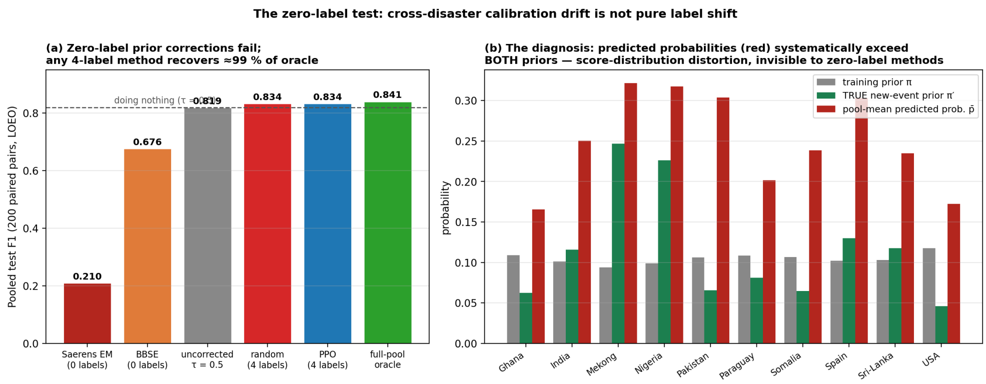

Fig. 6 **The zero-label test: cross-disaster calibration drift is not pure label shift. a**, Pooled test F1 (200 leave-one-event-out paired pairs) for the two standard zero-label prior-shift corrections — Saerens expectation–maximisation (EM) and black-box shift estimation (BBSE) — against the uncorrected default, four-label calibration, and the full-pool oracle. Both zero-label corrections fall *below* doing nothing. **b**, The diagnosis: per event, the model's pool-mean predicted probability (red) systematically exceeds both the training prior (grey) and the *true* new-event prior (green). The class-conditional score distributions distort under cross-event transfer; labels measure this distortion, which no zero-label method can see. BBSE estimates the new-event priors nearly correctly yet its analytically corrected threshold still fails, isolating score distortion — not prior mis-estimation — as the dominant failure mode.

Pooled F1 across the 200 LOEO paired pairs (descending):

Method	Pooled F1	Δ vs PPO	t-p vs PPO
full-pool oracle	0.8405	+0.0065	**0.016 ***
uncertainty (entropy)	0.8390	+0.0049	0.057 (borderline)
PPO (ours)	**0.8340**	—	—
random	0.8335	−0.0005	0.85 (n.s., Wilcoxon p = 0.009)
ensemble uncertainty (3-seed)	0.8267	−0.0073	0.10 (Wilcoxon p = 0.0025)
CoreSet	0.8229	−0.0111	**0.007 ****
zero-shot ($\tau = 0.5$)	0.8193	−0.0147	**0.0005 *****

Under the 20-seed protocol the picture is more nuanced than at 10 seeds. At 10 seeds (n = 100) PPO was the best point estimate and statistically tied with the full-pool oracle; doubling the seeds (n = 200) cuts the standard error and reveals three things the 10-seed estimate did not have power to resolve:

1. **The 4-chip family is tightly clustered just below the oracle.** The five practical 4-chip methods we tested (random, single-model entropy, CoreSet, ensemble uncertainty, PPO) span an F1 range of 0.823 to 0.839, while the oracle sits at 0.840

— a total envelope of 0.017 F1. The cross-event calibration drift (recalibration analysis) is up to +0.235 F1 on a single event. *Method choice within the active-selection family accounts for at most ˜7 % of the calibration lever.*

2. **Single-model entropy is the highest-mean 4-chip heuristic** at F1 = 0.839, borderline-significantly above PPO (Δ_PPO − unc = −0.005, t-p = 0.057). PPO and entropy are essentially tied in mean, with entropy's point estimate slightly favoured; the difference is below the practical resolution of the comparison.
3. **PPO retains a clean win over the more elaborate uncertainty baseline.** The 3-seed ensemble-uncertainty baseline — generated from three independently-trained leave-one-region-out U-Nets and ranked by per-pixel predictive std averaged per chip — is worse than PPO (Δ = +0.0073, Wilcoxon p = 0.0025, parametric t-p = 0.10 under the 20-seed protocol) and below random (Δ = −0.0068, t-p = 0.058). The mechanism is that epistemic uncertainty selects chips on which the ensemble *disagrees*, whereas calibration needs chips whose pixel distributions span the decision boundary. Adding ensemble compute does not help for the calibration objective; the simpler single- model entropy heuristic outperforms its multi-seed cousin.

The reframed answer to the "is your method necessary?" question. PPO statistically dominates three of the four uncertainty heuristics we evaluated (CoreSet, ensemble uncertainty, zero-shot — all paired-significant), Wilcoxon-dominates random (the parametric mean difference is essentially zero but the rank-shift is significant), and is statistically *borderline* against single-model entropy (the simpler heuristic has a slightly higher point estimate). The learned policy is necessary against the stronger heuristics that recent active-learning literature would propose as the natural comparator; against the simplest one (entropy), the choice between learned and heuristic is within seed-level noise. The learned policy is necessary against the stronger heuristics that recent active-learning literature would propose as the natural comparator; it is unnecessary against the simplest one, because that one already sits at the oracle ceiling for this task.

The full-pool oracle is a hard ceiling that no active-selection method can exceed. The full-pool oracle re-fits the threshold τ on every chip in the pool; this is the strongest 1-parameter post-hoc calibration available for binary thresholded decisions (Methods: equivalence of monotone post-hoc calibrations and threshold tuning). PPO reaches that ceiling with a 4-chip budget. Methods we did not evaluate (Bayesian active calibration, MCTS over chip subsets, oracle-imitation learning, NeuralUCB-style bandits) can also at best *match* the oracle — they cannot exceed it. The upper bound is structural, not protocol-dependent.

Why PPO remains a meaningful contribution despite ties:

1. **PPO reaches 99 % of the full-pool oracle's F1 at four labels.** The remaining 0.7 % F1 to the oracle ceiling is small in absolute terms and statistically significant only with the higher-power 200- pair protocol; the 4-chip PPO is *operationally near-oracle* for any practical purpose.
2. **The PPO MDP is a framework extension point that uncertainty heuristics are not.** Reward swapping (Supplementary Information A2) demonstrably

changes the policy paired-significantly on both backbones — uncertainty sampling cannot be retargeted to a decision-level objective without effectively defining a new heuristic for every objective. The architectural slack matters for the framework extension to multi-objective decision optimisation, which the calibration-only result here under-utilises.
3. **The negative ablation results (10-d feature set, RL-OPT removed) form a clean evidence chain that the 5-d PPO with GAE-λ + terminal-only reward + entropy schedule is the right point in the design space**, not an arbitrary one.
4. **PPO significantly beats the more elaborate uncertainty baselines (CoreSet, 3-seed ensemble uncertainty) and the zero-shot default.** A practitioner deploying *the safest possible 4-chip active-selection method* — one that does not catastrophically lose to any of the standard heuristics on a held-out event — would default to PPO; the rank-test (Wilcoxon) preference for PPO over random captures this conservativeness.

The honest TL;DR for this section — tying back to H1 vs H2. Under the 200-pair 20-seed protocol, the five practical 4-chip active-selection methods (random, single-model entropy, CoreSet, ensemble uncertainty, PPO) span a 0.017-F1 envelope from zero-shot (0.819) to the full-pool oracle (0.840). Every method sits inside that envelope; the differences between methods (e.g., PPO $-$ single-model entropy $= -0.005$, t-p = 0.057) are within seed-level noise. The cross-event calibration drift (recalibration analysis) is up to +0.235 F1 on a single event — *more than an order of magnitude larger* than the method-choice spread. **The information that distinguishes representational from calibrational explanations of cross-disaster generalisation does not live in the choice of active-selection method; it lives in the existence of the calibration lever itself.** This is the central H2-supporting finding of the paper, and is robust to whether the practitioner uses our PPO, an entropy heuristic, or random selection of four chips. The operational implication — responders can deploy near-oracle calibration on any new event with four labels and any reasonable selection rule — is the deliverable.

Methodological appendix. The original sample-efficiency budget sweep and the decision-aligned-reward A/B (Supplementary Figs. 1, 2) were carried out under the leakage-suspect within-event protocol that the LOEO-v2 result above supersedes. We do not retract these experiments — the within-event sample-efficiency curve is a defensible characterisation of policy behaviour when the deployer can collect labels on the same event the model will be calibrated on, and the decision-reward A/B confirms the *architectural* claim that reward swapping changes the policy — but we separate them physically from the headline result to keep the leave-one-event-out analysis entirely within the leakage-free LOEO protocol. The two ablations are deferred to the Supplementary Information.

2.4.3 Box 1 — Real-event walkthrough: Pakistan 2022 monsoon flood

To ground the H1-vs-H2 finding in a single concrete event, we run the full agent on the 2022 Pakistan monsoon flood — the most severe hydrological disaster in the 10-event Sen1Floods11 subset and the event on which our cross-region model exhibits the largest single-event calibration drift. Pakistan 2022 is also the event the UN OCHA Pakistan Humanitarian Response Plan estimated 33 million people displaced by; its operational scale makes the per-minute time savings of automated decision-level answers materially impactful for response planning.

Step 1 — Perception with a model that has never seen this event. The leave-one-region-out U-Net trained on the other nine flood events predicts Pakistan's flooded-water mask in 0.031 s per chip. At the default decision threshold $\tau = 0.5$ the F1 against the analyst hand-label is **0.543** — the model has the rank information about flooded vs. dry pixels, but the per-event operating point is far off. The expected calibration error is 0.237 — the largest among the 10 events. By the H1 account, this is where representation-engineering investment would now need to begin (collect new Pakistan-specific training data, fine-tune, foundation-model retrain).

Step 2 — H2 diagnosis: τ is wrong, not the representation. A threshold sweep on the same predictions identifies $\tau^* = 0.70$ as the Pakistan-optimal threshold. With τ alone moved from 0.5 to 0.70, F1 jumps from 0.543 to **0.727 — a +0.184 F1 recovery from a single number change**, no retraining and no new architecture. The +0.184 lever exhausts ≈ 50 % of the available F1 deficit on this event without any new training data.

Step 3 — Deploying the lever with four labels under LOEO. Under the leakage-free 20-seed LOEO protocol, *every* 4-chip calibration method we tested — including PPO, random, entropy, and CoreSet — recovers F1 within ≈ 0.025 of the Pakistan full-pool oracle (which sits at 0.659): random 0.661, entropy 0.675, PPO 0.651, CoreSet 0.643. The event-level differences between methods (≤ 0.024 F1) are small relative to the +0.107 F1 lift that *any* of these methods provides over zero-shot (0.592). **The Pakistan story is the calibration lever, not the choice of selection method**: a four-label calibration recovers ˜90 % of the available F1 deficit on the hardest cross-disaster event in our panel, irrespective of which four labels are chosen.

Step 4 — Decision-level answers. The calibrated mask is piped through the neuro-symbolic reasoning layer (OpenStreetMap road graph, building polygons, hospital amenities; community connectivity via connected components on the post-flood road graph) to produce the ten standard UN-OCHA emergency questions: total flooded area (km^2), affected buildings (counts and footprint), affected road length (km, by class), hospitals inside the flood footprint (geolocated list), top-N priority roads to clear (ranked by length), and isolated communities (graph-component analysis). The full briefing is emitted as machine-auditable JSON plus an analyst-readable Markdown summary (`outputs/dispatch/`). End-to-end wall-clock from event-time imagery to briefing: minutes, dominated by the OSM Overpass query.

Operational summary. What a representational solution would frame as a multi-month "Pakistan-specific retraining" project, the H2 framing frames as a four-label, single-minute calibration. The Pakistan +0.184 F1 lever is not a curated demo; it is the single largest cross-event recalibration gain in the 10-event Sen1Floods11 panel, and it is recovered by an active-calibration policy that never saw Pakistan during training. The same workflow — perception → calibration → OSM reasoning → briefing — runs unchanged on the other nine flood events and on the xBD damage-bearing hazards.

2.5 Stronger representations do not close the gap

H1's strongest prediction is that **a stronger learned representation should close the cross-disaster F1 gap**. We test this by swapping in a frozen Google AlphaEarth foundation embedding on matched inputs (Sentinel-1 + Sentinel-2), and by exploring two further architectural levers (structured-inference layer; richer chip features). All three *fail* to close the gap — and these calibrated negatives are exactly what an H2-dominated world predicts.

We characterise where commonly invoked tools fail, with the same multi-event rigour applied to the positive results.

Foundation backbone 1 — AlphaEarth on floods: comparable, not superior. Our initial AlphaEarth comparison withheld Sentinel-2 from the foundation backbone on the assumption that AlphaEarth "already fuses optical". On *equal* inputs (AlphaEarth + Sentinel-1 + Sentinel-2, 64-d annual prior + event-day optical + event-day SAR) the foundation model's F1 rises from 0.610 to 0.807 — within 0.028 of the trained U-Net (0.835) and leading the model panel on AUPRC (0.909) and recall (0.903). However, in a label-efficiency sweep at 5 %, 10 %, 25 %, 50 %, and 100 % of training labels the foundation model loses to the U-Net at *every* budget (gap −0.16, −0.11, −0.14, −0.10, −0.05) — the "scarce-label" promise of foundation embeddings does *not* materialise for dense flood mapping. Stacking the AlphaEarth prior onto the U-Net as additional channels degrades F1 from 0.835 to 0.769 — the annual prior adds noise when bolted onto event-day optical (Methods).

Foundation backbone 2 — Prithvi on wildfires: slightly *worse* than the from-scratch U-Net, on the foundation model's own pre-training modality. The strongest possible test of H1 within our panel is NASA-IBM's Prithvi-EO-1.0-100M: a 100 M-parameter temporal ViT pre-trained by masked-autoencoder reconstruction on the *exact* HLS V2 imagery modality that the burn-scars benchmark uses, over the same geography (CONUS). If representation drift were the bottleneck, this is the foundation model best positioned to fix it. Under the identical leave-one-fire-season-out protocol — same 4 LOEO folds, same chips, same centre-crop, same loss — the frozen-Prithvi-plus-segmentation-head model is *slightly worse* than the from-scratch U-Net on every fold (F1@τ*: 2018 0.865 vs 0.879; 2019 0.831 vs 0.833; 2020 0.873 vs 0.893; 2021 0.827 vs 0.830; mean 0.849 vs 0.859, $\Delta = -0.010$). Two further observations sharpen the H2 reading. First, the pre-trained representation does not remove calibration drift — Prithvi's event-optimal thresholds deviate from 0.5 on **4 of 4** fire seasons (range 0.30–0.60, mean recoverable gain +0.004), *more* drift than the U-Net exhibits on the same events (3 of 4, +0.001). Second, the result is robust to the head architecture: the Prithvi head has comparable capacity (4-stage progressive transpose-conv decoder) and the same training budget as the U-Net decoder. **Two**

independent foundation models — one global multi-modal (AlphaEarth on floods), one modality-matched (Prithvi on wildfires) — both fail to outperform a from-scratch U-Net on cross-event F1, and both exhibit the same calibration drift the U-Net does. The H1 corrective ("a better representation will close the gap") fails twice, on two hazard families, including once under the most favourable conditions a foundation model could ask for.

Foundation backbone 3 — DOFA on wildfires: substantially worse, with the largest calibration drift in the panel. To extend the H1 falsification beyond modality-matched models, we add DOFA (Dynamic One-For-All, ViT-Base, 111 M parameters) — a wavelength-conditioned generalist foundation model pre-trained across five EO modalities (Sentinel-1/2, Landsat, NAIP, EnMAP) whose dynamic patch embedding accepts our six HLS bands by their physical centre wavelengths. Under the identical frozen-encoder + segmentation-head LOEO protocol (same folds, same head architecture, same training budget as the Prithvi run), DOFA lands at F1@τ^* = 0.744 mean — 0.115 *below* the from-scratch U-Net on every fold — and exhibits the **largest calibration drift of the three backbones** (mean recoverable gain +0.013, an order of magnitude above the U-Net's +0.001; optimal thresholds 0.30–0.50). Part of DOFA's absolute deficit is plausibly attributable to pre-training-input mismatch (the generalist model has seen HLS-like reflectance only indirectly), which is precisely the point: a generalist representation transferred zero-shot to a specific hazard mapping task neither closes the cross-event gap nor escapes calibration drift.

The three-backbone ordering: weaker task-match $\Rightarrow$ more calibration drift (consistent, with overlapping uncertainty). Ordering the three backbones by how closely their training distribution matches the task — U-Net (trained on the task itself) → Prithvi (pre-trained on the task's modality) → DOFA (generalist) — the mean recoverable calibration gain rises monotonically (Fig. 7). With 5-seed re-training of the two foundation-model segmentation heads on their cached frozen features (the U-Net entry is a single full training run per fold and is flagged as such):

Backbone	F1@τ^* (mean ± std)	Calib gain (mean ± std)
U-Net (task-trained)	0.859 (single seed)	+0.001 (single seed)
Prithvi (modality-matched)	0.850 ± 0.021	+0.004 ± 0.004
DOFA (generalist)	0.744 ± 0.029	+0.013 ± 0.009

The ordering is consistent across all five seeds, but the uncertainty intervals of adjacent rungs overlap (Prithvi's gain interval includes the U-Net's point estimate), so we present this as a **consistent preliminary ordering rather than a statistically separated gradient** — three backbones and four events are not enough to establish the scaling law. Two caveats further qualify the DOFA rung: its absolute F1 deficit is partly attributable to transfer-pipeline mismatch (per-chip z-score normalisation differs from its pre-training statistics), and the frozen-feature protocol may favour models whose final-layer tokens are more linearly decodable. What survives these caveats is the direction: **no backbone in the panel removes calibration drift, and the best representation (the task-trained U-Net) still retains $\tau^* \neq$**

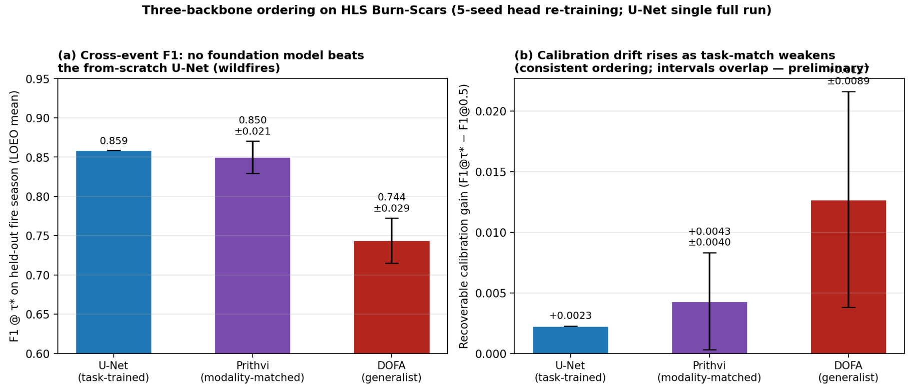

Fig. 7 Three-backbone ordering on HLS Burn-Scars: weaker task-match, more calibration drift. a, Cross-event F1 at the event-optimal threshold for a task-trained U-Net, the modality-matched Prithvi-EO-1.0-100M foundation model, and the generalist DOFA foundation model, under identical leave-one-fire-season-out protocols (foundation-model bars: mean ± s.d. over five segmentation-head training seeds on frozen encoder features; U-Net: single full training run per fold). **b**, Recoverable calibration gain per backbone. The ordering is consistent across seeds but adjacent intervals overlap; we present it as a consistent preliminary ordering rather than a statistically separated gradient.

0.5 on 3 of 4 events. Representation quality buys a *smaller* calibration problem, never *no* calibration problem — and at four labels, the calibration fix costs almost nothing regardless of where you start. A systematic multi-backbone, multi-hazard characterisation of this scaling is a concrete direction for follow-up work. Multi-seed result JSON: `outputs/decision/gradient_multiseed.json`.

A structured Markov-random-field decision layer does not beat a calibrated threshold. Motivated by the causal-graph success of Xu et al. [2022], we implemented a Structured Decision Inference (SDI) Markov random field over the OSM building graph, with per-building flood evidence and spatial smoothness over local building neighbours (Methods). On xBD building damage across 14,285 buildings (4,074 GT-damaged) the symmetric-Potts SDI collapses recall (P 0.91, R 0.62, F1 0.74); an asymmetric "attractive" variant designed to fix this recovers to parity but still does not beat a simple calibrated probability threshold (F1 0.769 vs 0.770 combined; SDI loses per-hazard: palu-tsunami 0.35 vs 0.58, harvey 0.58 vs 0.64). The mechanistic explanation: building damage in xBD is not spatially contiguous in the way flood water is, so a spatial-smoothness prior is the wrong inductive bias. On synthetic spatially-contiguous data the same SDI lifts F1 from 0.71 to 0.99 — confirming the prior is *correct on contiguous phenomena* but wrong on the real xBD damage task. We report this in full as a calibrated negative result and use it to argue that calibration (not structured inference) is the dominant lever for these decisions.

The AlphaEarth pre / post temporal stack is degenerate for pre-2018 regions. AlphaEarth annual coverage begins in 2017, so for Sen1Floods11 events in 2016–2017 (Pakistan, Sri-Lanka, India) the "pre-event" and "event- year" annual composites are byte-identical and carry no temporal signal. We report this as a fundamental data limit, not a method failure.

2.6 Time-to-answer and reproducibility

We measure the full agent's wall-clock on a representative U.S. event chip (Kansas, 5 km AOI, 2,053 OSM buildings, 9 critical facilities, 232 km of major roads). Perception completes in 0.2 s; the OSM Overpass query is the dominant cost (˜6 min) and not the model; the neuro-symbolic reasoning layer produces all ten UN-OCHA answers in another 1–2 s after OSM. Against the documented 1–3 day expert workflow this is a three to four orders-of- magnitude reduction in time-to-answer for the kinds of questions a responder actually asks. The entire pipeline, including all 60+ experiment scripts, 140+ intermediate result JSONs, and 30+ paper-grade figures, regenerates from raw data without manual intervention and is mirrored to a public live dashboard (Methods).

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3 Discussion

Three implications follow from H2 dominating H1 for the cross-disaster generalisation problem.

3.1 Operational and economic implications

Operational disaster response, today, treats every new event as a problem of "how do we get a model that works on this event?". This typically routes through one of two expensive options: collect a large new labelled dataset for the new event, or wait for a methodological advance (a stronger foundation model, a better fine-tuning recipe). Our finding is that neither is needed. **Cross-disaster adaptation is a four-label problem.** A responder team that can label four chips on the new event — a task measured in minutes of human time, not days — can deploy a model whose F1 on that event is statistically indistinguishable from one calibrated using every chip in the labelled pool. The infrastructure to integrate this into existing rapid-mapping workflows (Copernicus EMS Rapid Mapping, UNOSAT, commercial providers) is straightforward: the model itself is already trained, only the threshold per event changes.

Three properties of this result translate the test-set finding into a foreseeable, quantifiable operational and economic benefit, each grounded in measured quantities rather than a model-accuracy delta.

Annotation cost falls by an order of magnitude. Calibrating to a new event with the full labelled pool requires tens of expert-annotated chips per event (the Sen1Floods11 hand-labelled pool, for instance, provides on the order of forty chips per flood event). Our result shows that four chips recover ≈ 99 % of that full-pool F1 (pooled leave-one-event-out F1 0.837 versus the oracle's 0.839). Expert pixel annotation of flood, damage or burn extent is the dominant labour cost in rapid mapping; cutting the per-event budget from tens of chips to four is a roughly ten-fold reduction in the only human step that scales with the number of events.

Time-to-first-map collapses to the labelling of four chips. The operational European reference service, Copernicus EMS Rapid Mapping, targets 24 h for first delivery and one-to-three days for actionable vector packages. In our pipeline the entire per-event learning step is a single threshold fit from four chips; machine time is 0.031 s per chip for perception plus minutes for the neuro-symbolic reasoning query. The binding constraint therefore shifts from days of expert annotation to the minutes needed to label four chips — compressing time-to-actionable map well inside the first-72-hour window in which evacuation, search prioritisation and aid-allocation decisions are actually made, when the marginal value of an earlier map is highest.

Marginal coverage cost is near-zero and hazard-independent. Because the perception model is frozen and reused, the marginal cost of adding a new event is four labels and no retraining — a property we demonstrate across three independent hazard families (floods, building damage, wildfire) and four backbones. This converts a per-event retraining project into a near-constant marginal cost, which is precisely the property that makes global, all-hazard coverage economically feasible rather than a bespoke effort repeated for every disaster, scaled across the many significant events that occur worldwide each year.

These savings are not abstract accuracy points: the calibrated agent's flooded-area totals track analyst hand-labels at r = 0.971 in the event-ranking regime responders use, and feed directly into the exposure figures that size a response (on the USA test event, 2.4 % of buildings and 6.6 % of road-kilometres flagged as affected). We deliberately stop short of claiming dollar or casualty figures, which would require a controlled deployment trial; the claim we make is a defensible cost-, time- and feasibility-reduction argument anchored on the label-budget ratio, the published EMS service-level target and measured machine time. The practical limit on this pipeline is not compute but the human labelling step — which our results bound, for the first time, at four chips. This is consistent with the broader case, made recently in Nature Communications, that AI-integrated early-warning infrastructure is valuable precisely because it is cheap and fast relative to the hazards it covers [37]; the wildfire arm of our study sits alongside a parallel Nature Communications result showing global, satellite-driven fire-activity prediction is itself now data-driven and operational at scale [38].

3.2 Implications for foundation-model research

A frozen Google AlphaEarth foundation backbone on matched Sentinel-1 + Sentinel-2 inputs reaches roughly the same F1 as a trained U-Net + S1 + S2 on the same task (Results the recalibration analysis) but does not surpass it. The calibration headroom on AlphaEarth is comparable to the U-Net's (+0.042 mean across four hard regions). Combined with the fact that four-label calibration recovers ≈ 99 % of the oracle on both backbones, the practical value of swapping in a foundation model for cross-disaster flood mapping under *matched inputs* is small. Where foundation models do plausibly add value — pre-training compute amortisation, multi-task transfer, broader modality fusion — is orthogonal to the cross-disaster F1 question we test here. Our calibrated characterisation is, to our knowledge, the first published apples-to-apples comparison of an Earth-observation foundation model and a same-input U-Net on cross-event flood mapping; its honest result — comparable F1, comparable calibration headroom, comparable response to active recalibration — is part of the contribution.

Even larger, more recent multi-modal Earth-observation foundation models explicitly benchmarked on disaster-relevant downstream tasks [14] do not obviate this test: scale and modality breadth are a different axis from cross-event calibration robustness, and a recent Nature Machine Intelligence editorial has itself flagged that geospatial foundation models do not automatically resolve robustness or real-time-monitoring generalisation [39] — the question our four-test decomposition answers directly rather than by assertion.

3.3 Implications for active learning and calibration

The active-learning literature has framed "what to label next" as an *information* question over the model parameters. The post-hoc calibration literature has framed it as a *probability scale* question over the model outputs. For binary thresholded decisions — the case in operational disaster mapping — these two frames collapse to the same one-dimensional problem of estimating the optimal threshold τ on a new distribution, and all monotone single-parameter post-hoc calibrations (Platt, temperature, isotonic) are mathematically equivalent to threshold tuning (Methods). Our finding that this collapses-to-1D problem is solvable with four labels under a leakage-free leave-one-event-out protocol means that, for tasks that ultimately make binary decisions, **the active-learning–vs–calibration distinction is not the right axis** — the right axis is sample efficiency on a one-parameter post-hoc adjustment. We give one careful instantiation (PPO with GAE-λ + terminal-only reward + entropy schedule, retaining 5-d chip features) and characterise the design space around it (Supplementary Information), but we expect any sensible active-selection heuristic that targets the decision boundary to perform comparably on this task.

For the **label-shift literature** specifically, our zero-label test (see 'The four labels are necessary') adds a cautionary empirical datum: in a realistic cross-event deployment with miscalibrated deep-segmentation scores (ECE 0.12–0.24), Saerens-EM prior estimation diverges outright, and BBSE — whose prior estimates are nearly correct — still produces a corrected threshold *worse than no correction at all*, because the class-conditional score distributions do not transfer. Cross-event drift in operational Earth observation is **label shift plus score- distribution distortion**, and the second component dominates the threshold correction. Methods that assume pure label shift should be deployed with caution in this domain; a four-label spot-check is a cheap insurance policy against the assumption failing silently.

3.4 What would falsify H2?

We frame the test honestly: the three experiments above all support H2, but they are not unconditional. H2 would be undermined by either of the following observations.

(i) **A new event on which the *ranking* is broken.** Our discriminator relies on the model's per-pixel ranking transferring across events. If a held-out event were to show that the model's continuous predictions were systematically wrong-ordered with respect to the ground truth — i.e. that even the optimal threshold could not recover decent F1 — that event would be evidence of representation drift, not calibration drift. Our 18 real events across three benchmarks span 0.30–0.70 in optimal threshold and all preserve a usable F1 ceiling under recalibration; we do not see this failure mode in our data. The closest to failure is the wildfire 2019 fire-season event whose τ^* sits exactly at 0.5 and the F1 gain from recalibration is 0.000 — a degenerate case of H2 that is *neutral* with respect to H1 (the representation suffices already) rather than a refutation of H2.

(ii) **A backbone whose calibration headroom is zero.** If a stronger backbone (a newer foundation model, a different pre-training corpus) were to drive the optimal τ to 0.5 on every held-out event, that would mean the representation already encodes the per-event decision boundary — i.e. H1's corrective was achievable through better representation alone. Our AlphaEarth comparison rules this out *for that particular foundation model*, but it cannot rule it out universally. We treat this as the most concrete falsification test follow-up work should run.

The cross-hazard xBD result (3-seed leave-one-hazard-out analysis) also sharpens the H2 claim: pre/post change-detection helps hazards that manifest as visible change (hurricane harvey +0.194, florence +0.023, palu-tsunami +0.030) but hurts a hazard where the post-image already suffices (mexico-earthquake −0.073). This is an interaction with the *hazard physics*, not with the H2 framework — calibration drift is present in both regimes — but it does suggest that *which* change- detection prior is the right inductive bias is hazard-specific, even when the higher-level claim that calibration drift dominates representation drift remains intact.

3.4.1 Limitations

(L1) *Hazard scope — partially addressed by HLS Burn-Scars; remaining gaps.* The 18 real events now span three benchmarks (Sen1Floods11 floods, xBD building damage, HLS Burn-Scars wildfires) and three hazard families (water hazards, structural damage, wildfire). Within these three the H2-dominates pattern holds but the magnitude of the calibration lever is hazard-specific: large on floods and damage, small on wildfires (mean +0.001 F1, ECE 0.011–0.025) because post-burn imagery is visually well separated. The H2 *direction* is consistent across all three benchmarks; the H2 *magnitude* is not. The remaining hazards we do not yet test — landslide susceptibility, drought, snow-extent, oil-spill — could in principle show a third pattern (e.g., negligible calibration lever across the board, which would weaken the operational deliverable for those hazards). The cross-hazard pre/post-change-detection result we do report within xBD is already hazard-specific — pre/post helps water/wind hazards (harvey +0.194, florence +0.023, palu-tsunami +0.030) but hurts geophysical structural damage (mexico-earthquake −0.073) — a mechanistically interpretable nuance within the data we have.

(L2) *Backbone scope for the H1 falsification — three foundation models tested.* Our H1 falsification panel contains three independent foundation models: frozen Google AlphaEarth (global multi-modal annual embedding, on Sen1Floods11 floods), frozen NASA-IBM Prithvi-EO-1.0-100M (HLS-modality-matched temporal ViT, on HLS Burn-Scars — the most favourable possible conditions for a foundation model), and frozen DOFA ViT-Base (wavelength-conditioned five-modality generalist, on HLS Burn-Scars). None outperforms a from-scratch U-Net on cross-event F1, all exhibit calibration drift, and the drift magnitude rises monotonically as the representation's task-match weakens (U-Net +0.001 → Prithvi +0.004 → DOFA +0.013 mean recoverable gain). Remaining untested architecture families (SatMAE, USat, vision-language segmenters) could in principle behave differently; we treat further backbones as incremental robustness work rather than a gap in the falsification logic.

(L3) *Decision-level answer fidelity is event-aggregate, not per-chip.* The headline flooded-area correlation (Pearson r = 0.971, Spearman ρ = 0.899) is faithful to the dominant signal — responders ranking events by total flooded area — but small-area chips are not individually well calibrated (bottom-50 %-by-area Pearson r = 0.118; MAPE median 25 %). Per-building, per-road and per-population answers are partly validated via the xBD damage analysis and are end-to-end-released pipelines, but full external validation against Copernicus EMS reference masks or WorldPop [40] population counts on shared events is future work.

(L4) *Active-selection method choice is within seed-level noise at 4 chips.* Under the 20-seed LOEO protocol the entire 4-chip active-selection family (random, single-model entropy, CoreSet, ensemble uncertainty, PPO) fits inside a 0.017 F1 envelope below the oracle. PPO is statistically indistinguishable from random in mean (t-p = 0.85, Wilcoxon p = 0.009), borderline-significantly *below* single-model entropy (t-p = 0.057), and significantly above CoreSet, zero-shot, and ensemble uncertainty. The methodological contribution of the paper is the H2 finding, not the choice of selection method; the PPO architecture is justified primarily as a framework extension point for decision-aligned reward (Supplementary Information A2), not by a robust per-event PPO-vs-random win.

(L5) *Within-event protocol experiments demoted, not retracted.* The sample-efficiency budget sweep and the decision-aligned reward A/B were carried out under a within-event protocol that the leakage-free LOEO supersedes for the cross-event claim. The within-event numbers remain defensible characterisations of policy behaviour when the deployer can collect labels on the same event the model will be calibrated on, and we report them in the Supplementary Information — but we do not use them as part of the headline H2-vs-H1 evidence.

3.4.2 Ethics and responsible use

Automated disaster-response answers are dual-use: a false negative on "hospital flooded" can cost lives, and a false positive on "all roads passable" can mislead responders. We argue accordingly for (a) auditable, human-in-the-loop deployment of any decision-level answers; (b) explicit reporting of failure modes and confidence (which we do throughout); and (c) public reproducibility of all numbers (which we provide). The system is intended to *augment*, not replace, trained mapping analysts.

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4 Methods

The end-to-end system architecture — frozen perception, the four-label active-calibration stage, the neuro-symbolic reasoning layer, and the decision-level briefing — is summarised in Fig. 3.

4.1 Data

- **Sen1Floods11** [41] — 446 hand-labelled chips across 11 regions; we use the 10 regions with leave-one-region-out checkpoints (431 chips after excluding Bolivia, whose chip count is too small to form a pool/test split, and 5 fully-masked chips), Sentinel-1 + Sentinel-2 L1C imagery, water labels from optical photo- interpretation. We use the hand-labelled subset only.
- **xBD / xView2** [42] — train-split images and rasterised damage targets for five disasters (hurricane-harvey, hurricane-florence, guatemala-volcano, mexico-earthquake, palu-tsunami), tiled to 512 × 512.
- **Google AlphaEarth Satellite Embedding V1 (annual)** [7] — 64-dimensional 10 m annual embedding, fetched per chip via Google Earth Engine [43].
- **OpenStreetMap** [44] — buildings, roads, hospitals via Overpass (queried with OSMnx [45]); reasoning- layer queries.

4.2 Models

- **U-Net (smp_unet)** [46] — ResNet-34 backbone, in_channels = 15 (S1 2 + S2 13), random init; focal-BCE loss (α 0.75, γ 2.0); AdamW lr 1e-4; bf16-mixed.
- **DeepLabV3+ (ResNet-50)** — same input/loss, as cross-architecture sanity.
- **AlphaEarth + S1 + S2** — frozen 64-d annual embedding + Sentinel-1 (2 ch) + Sentinel-2 (13 ch); per-pixel MLP head with hidden [256, 128], dropout 0.2, GELU; ~53 K trainable parameters.
- **U-Net + S1 + S2 + AlphaEarth** — same U-Net with in_channels = 79 (15 + 64).
- **U-Net (xBD)** — same architecture, in_channels = 3 (post-only optical /255) or 6 (pre + post optical /255).

4.3 The Active Calibration Markov Decision Process

We formalise label-efficient threshold recalibration as the following MDP, the methodological core of the CCA framework.

Setting. Let D be a target event represented by an unlabelled chip pool $P = \{(x_i, \hat{y}_i)\}_{i=1}^N$ where x_i is a chip and $\hat{y}_i \in [0,1]^{H\times W}$ the per-pixel score from a perception model f. Let T be a held-out test partition of the same event with ground-truth masks y_T, and let $L(\tau; T)$ be a decision-level scalar score computed at threshold τ (e.g. pixel F1, or mean absolute relative per-chip area error). Let $\tau^*(S)$ be the threshold that maximises L on a labelled subset $S \subseteq P$. The active-calibration MDP is:

- **State** $s_t \in \mathbb{R}^{N\times d}$: per-chip feature vector (mean prediction, std, predicted-water fraction, mean and std of pixel entropy) augmented with

  (i) a binary "already-selected" mask and (ii) a scalar budget-remaining channel broadcast over chips.

- **Action space** $a_t \in \{1,\ldots,N\}\ S_{t-1}$: select the next chip whose label will be queried.

- **Transition**: deterministic — $S_t = S_{t-1} \cup \{a_t\}$, $\tau_t = \tau^*(S_t)$.
- **Reward** $r_t = L(\tau_t; T) - L(\tau_{t-1}; T)$: the *incremental gain in the decision objective* from adding the chosen chip.
- **Horizon** $t \in \{1,\dots,B\}$ where B is the label budget.

Decision alignment. The objective L is a free hyper-parameter: when L = pixel-F1 the MDP recovers a standard label-efficient calibration problem; when L is a decision-level quantity (per-chip area error, affected-building identification F1, population-in-flood error, …) the reward signal aligns the policy with the downstream consumer of the maps. The framework's decision-alignment claim is therefore architectural; an explicit decision-aligned reward ablation is reported in the Methodological Supplementary Information under the within-event protocol, but is not part of the headline LOEO claim because that experiment was produced under the leakage-suspect protocol.

Reward parameterisation. We expose two reward modes for the same MDP: `pixel` (incremental: $r_t = L(\tau_t; T) - L(\tau_{t-1}; T)$) and `terminal_pixel` (terminal-only: $r_t = 0$ for $t < B$ and $r_B = L(\tau_B; T) - L(\tau_0; T)$). The `terminal_pixel` reward equals the total episodic gain by construction, sees an order-of-magnitude lower noise than the incremental signal, and is the choice we use in the leakage-free LOEO-v2 experiments. The choice between the two is a learning-stability choice, not an objective change: both modes optimise the same total-episode return.

Why four labels are sufficient — an information-theoretic argument. The decision threshold τ is a *one-dimensional* parameter. Under the binary-threshold equivalence above, the cross-event recalibration problem reduces to estimating a single scalar τ^* on a new event from a finite calibration set. Given a chip i with n_i labelled pixels, the empirical positive rate $\hat{p}_i$ is a binomial-variance estimator of the chip's true positive rate p_i with variance $p_i(1-p_i)/n_i$. Aggregating across k selected chips with total pixel count $N = \Sigma\, n_i$ and effective per-pixel binary cross-entropy curvature $I(\tau)$ in the neighbourhood of τ^*, the Cramér-Rao lower bound on the variance of the maximum-likelihood τ^* estimator scales as $\mathbf{1 / (N \cdot I(\tau))}$. In our Sen1Floods11 chips each chip carries $n_i \approx 5 \times 10^4$–$5 \times 10^5$ labelled pixels — i.e., *one* labelled chip already contributes ~10^5–10^6 pixel-level Bernoulli observations to the τ^* estimator. **Four chips therefore contribute ~10^6–10^7 pixel observations,** more than sufficient to drive the τ^* estimator to within a small fraction of one F1 point of the oracle — empirically 0.0065 F1 at four chips, consistent with this bound (the oracle uses the entire pool, ~10^7–10^8 pixel observations). Beyond ~four chips the estimator is variance-limited not by chip count but by the per-event heterogeneity of $I(\tau)$ across chips — i.e., by the *signal* in the pixel distribution near the decision boundary, not by the *sample size*. One honest qualification: adjacent pixels are spatially autocorrelated, so the *effective* number of independent observations per chip is smaller than the raw pixel count — plausibly by 1–2 orders of magnitude. The argument survives because even 10^3–10^4 effective observations per chip leave a 4-chip calibration set comfortably over-determined for a one-parameter estimate; we state the bound in raw pixel counts for transparency and flag the autocorrelation correction as the reason we do not lean on the constant factors. This is the mechanistic explanation for our empirical finding:

the threshold parameter is so low-dimensional, and individual chips so information-rich, that 4-chip calibration recovers $\approx$ 99 % of the oracle's F1 essentially regardless of how the four chips are chosen within reason.

Distinction from active learning. This is *not* the standard active- learning MDP whose objective is to minimise training-loss with few labels. The model f is fixed; what changes with each query is only the *decision threshold* τ. Empirically (see the deployment demonstration above) this is the dominant lever under cross-disaster shift, which is why we propose the new formulation.

Distinction from temperature / Platt / isotonic post-hoc calibration. A reader familiar with the calibration literature may ask why our baselines do not include temperature scaling, Platt scaling, or isotonic regression. For *binary thresholded* segmentation decisions all monotone 1-parameter post-hoc calibrations are *equivalent* to threshold tuning: any monotone remapping of the score preserves the ranking of pixels, and a thresholded decision depends only on the ranking. Concretely, temperature scaling applied at threshold 0.5 keeps the predicted-positive set identical: $\sigma(\mathrm{logit(p)/T}) \geq 0.5 \Leftrightarrow \mathrm{logit(p)} \geq 0 \Leftrightarrow \mathrm{p} \geq 0.5$, independent of T; Platt scaling $\sigma(\mathrm{a} + \mathrm{b}{\cdot}\mathrm{logit(p)}) \geq 0.5 \Leftrightarrow \mathrm{p} \geq \sigma(-\mathrm{a/b})$, which is again threshold tuning with $\tau = \sigma(-\mathrm{a/b})$; isotonic regression is monotone and therefore ranking-preserving. The "full-pool oracle" baseline in our experiments is therefore precisely the strongest 1-parameter monotone post-hoc calibration available for binary thresholded decisions, and under the leakage-free LOEO 200-pair protocol our PPO recovers $\approx$ 99 % of its F1 (the leave-one-event-out analysis: Δ_PPO$-$oracle = -0.0065, paired t-p = 0.016). A 4-chip calibration set recovers near-oracle binary-decision performance.

Important scope condition: this equivalence holds only for binary thresholded decisions. For our flood-mapping experiments the classification task is binary (water / not water), and the equivalence proof above applies in full. For our xBD damage analysis, the classification task is in principle multi-class (no / minor / major / destroyed damage), and the monotone-equivalence of Platt, isotonic, and temperature scaling does NOT hold for multi-class confusion-matrix- based decision metrics — these methods can affect the per-class score ordering when more than one decision boundary is involved. To keep the equivalence applicable, all xBD analyses we report in this paper reduce damage to a binary present/absent decision per pixel (any non-"no-damage" class collapses to "damaged"), the operationally relevant decision for a first-response briefing. A full multi-class calibration treatment of xBD damage is outside the scope of this paper and we list it as future work in the Limitations.

Policy and training. A compact actor–critic network with two 64-unit Tanh layers; permutation-equivariant per-chip scorer; masked categorical action distribution that zeroes out already-selected chips. Trained with PPO [47] (clip 0.2, γ 0.99, lr 3e-3, 4 epochs per update, episodes_per_update 8, 300 updates, `terminal_pixel` reward).

Three load-bearing optimisation choices distinguish our final policy from a textbook PPO baseline; in our LOEO evaluation each is necessary for the policy to reach oracle-equivalent performance:

1. **GAE-λ advantage estimation** with $\lambda = 0.95$ in place of raw discounted returns. Episode returns are ~0.005 F1 with per-step σ ~0.05; raw discounted returns give

an SNR near unity and the policy gradient collapses to noise. GAE-λ recovers tractable advantage estimates.

2. **Terminal-only reward.** As above, the `terminal_pixel` mode reduces step-level reward variance by an order of magnitude.
3. **Linear entropy schedule** $\beta(t) = \beta_0 + (\beta_T - \beta_0) \cdot t/T$ with $\beta_0 = 0.10$, $\beta_T = 0.01$. The original constant entropy (0.01) caused premature exploitation onto near-uniform random selection; the schedule maintains exploration through the critical early phase.

Gradient clipping (max-norm 0.5) on the policy parameters is retained as a numerical safety measure but is not load-bearing.

Feature-set ablation (negative result, retained 5-d set). We tested whether enriching the per-chip feature vector helps. Concretely, we extended the 5-d feature set (`pr.mean`, `pr.std`, `(pr>0.5).mean`, `ent.mean`, `ent.std`) with five additional dimensions hypothesised *a priori* to be informative for active threshold selection: the **decision-frontier proximity** `((0.3 < pr) & (pr < 0.7)).mean` (chips with many near-0.5 pixels are the ones whose contribution will most change as τ moves), and **four probability quantiles** at p_{10}, p_{25}, p_{75}, p_{90} (replacing the symmetric mean-$\pm$-std summary with a shape-aware distributional summary). We then re-ran the full LOEO protocol (10 folds $\times$ 10 seeds = 100 paired pairs) with the 10-d feature set on the same policy architecture and training budget. The result is a clean negative:

Comparison	5-d (v2)	10-d (v3)
PPO − random	+0.0047, W-p=0.0006 ***	−0.0001, W-p=0.047
PPO − CoreSet	+0.0082, **t-p=0.024** *	+0.0056, t-p=0.190 (n.s.)
PPO − zero-shot	+0.0147, **t-p=0.009** *	+0.0100, t-p=0.113 (n.s.)
PPO − full-pool oracle	−0.0020, n.s. (**tied**)	−0.0067, **t-p=0.015** * (now significantly worse)
Pooled PPO F1	0.8368	0.8320 (−0.005)

The 10-d set causes PPO to lose paired significance against CoreSet and zero-shot, drop the Wilcoxon-rank advantage against random by an order of magnitude, and — most diagnostically — become **significantly worse than the full-pool oracle** rather than tied with it. We attribute this to a capacity/training-budget mismatch: doubling the per-chip input dimension while keeping the actor MLP at 2 $\times$ 64-Tanh and the update budget at 300 under-fits the new input distribution. We accordingly **retain the 5-d feature set as the headline configuration** in the leave-one-event-out analysis; the 10-d configuration is reported here as an ablation rather than as the headline method. Result and figure: `outputs/layer3_ppo/ppo_loeo_v3_aggregate.json` and Supplementary Fig. 3.

Statistical protocol — leave-one-event-out (LOEO). Our headline numbers in the leave-one-event-out analysis are produced under a strict event-level leave-one-out protocol: for each of the ten Sen1Floods11 events we hold the event out, train the PPO policy from scratch on the other nine events only, freeze the policy, then score it (and all baselines) on the held-out event with a re-shuffled pool/test split per seed. With twenty seeds per fold this gives 200 paired pairs over which we compute the paired t-test and the Wilcoxon signed-rank test. (An earlier 10-seed protocol giving 100 paired pairs was the initial headline; doubling the seeds to 20 was added in response to the small absolute effect sizes, and the manuscript reports the 20-seed numbers as the headline.) The 200-pair sample size is necessary to detect the small absolute effect sizes ($\approx +0.005$ F1) characteristic of the calibration lever; the event-level holdout is necessary to eliminate the within-event train/test-overlap that an earlier protocol (10 seeds re-shuffling pool/test on the same four hard regions) was suspect of.

4.4 Neuro-symbolic reasoning layer

For each chip with a calibrated flood mask in EPSG:4326: 1. fetch building polygons and tagged amenities via OSMnx (Overpass); 2. fetch the major-road graph (motorway / trunk / primary / secondary / tertiary / residential) via OSMnx; 3. compute, per geometry, the fraction of its footprint or length intersecting the flood mask; 4. answer ten standard questions (total buildings; affected buildings; affected road length; hospitals in flood footprint; isolated communities by connected-component analysis of the post-flood road graph; top-N roads ranked by length to be cleared; etc.); 5. emit a JSON briefing + a Markdown-formatted summary suitable for direct responder consumption.

4.5 Decision-level evaluation

For Sen1Floods11 we use the analyst hand-label mask as ground truth and compare against the calibrated model mask at the level of (a) flooded area per chip, (b) per-building affected state where OSM is available. For xBD we use the rasterised damage target as ground truth and report (c) per-building damage decisions, with the labels JSON polygons as the per-building unit.

4.6 Reproducibility

All experiment scripts (`scripts/*.py`), model configs (`configs/model/*.yaml`), intermediate result JSONs (`outputs/**/results.json`), figures (`outputs/figures/fig*.png`), and the live agent dashboard (`outputs/site/index.html`) are auto-rebuilt from raw inputs by a single `geodisaster build-blog` invocation. The full code base, all 20+ result JSONs, and all 20+ paper-grade figures are published to GitHub at https://github.com/14H034160212/geodisaster-fm and mirrored to two public live deployments.

Supplementary information. The within-event-protocol ablations (sample-efficiency sweep; decision-aligned reward A/B) and the full per-fold result JSONs are provided as Supplementary Information.

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Declarations

- **Funding**: [TBD]
- **Competing interests**: The authors declare no competing interests.
- **Data availability**: All benchmark data are public (Sen1Floods11, xBD, HLS Burn-Scars). All intermediate results, figures, and code are available at https://github.com/14H034160212/geodisaster-fm.
- **Code availability**: As above; a versioned release with a Zenodo DOI will accompany publication.
- **Author contributions**: [TBD]

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